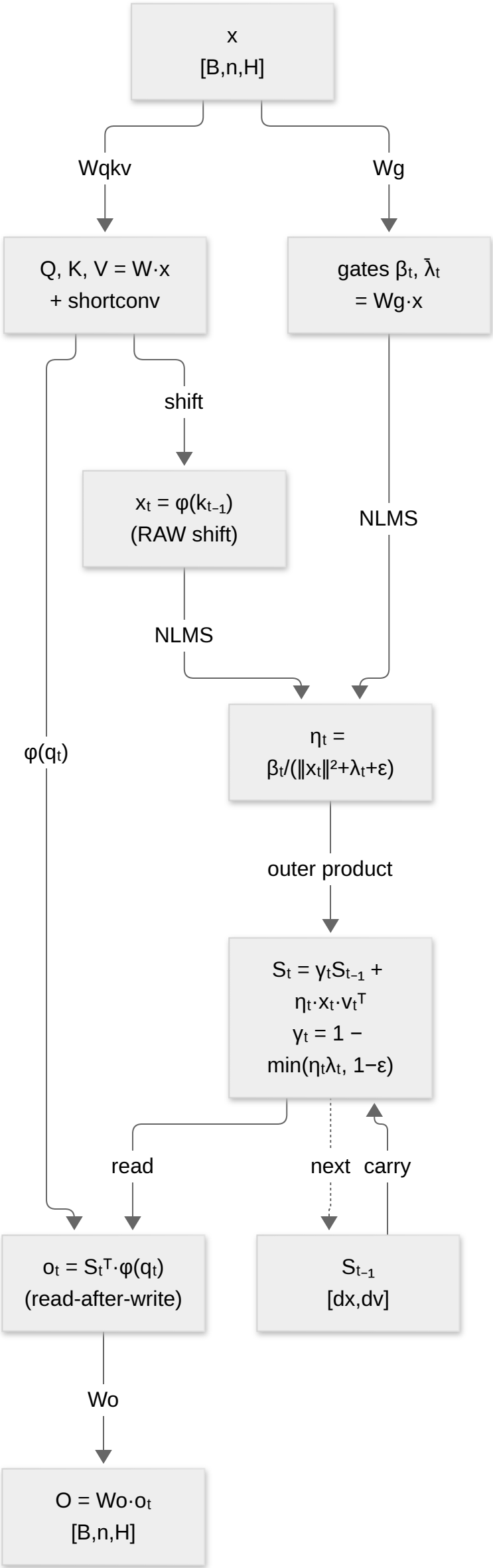

x
[B,n,H]
Wqkv
Wg
Q, K, V = W·x
+ shortconv
gates β_t, $\bar{\lambda}_t$
= Wg·x
shift
$x_t = \varphi(k_{t-1})$
(RAW shift)
NLMS
NLMS
$\varphi(q_t)$
η_t =
$\beta_t/(\|x_t\|^2+\lambda_t+\varepsilon)$
outer product
$S_t = \gamma_t S_{t-1}$ +
$\eta_t \cdot x_t \cdot v_t^T$
γ_t = 1 −
min($\eta_t\lambda_t$, 1−ε)
read
next
carry
$o_t = S_t^T \cdot \varphi(q_t)$
(read-after-write)
S_{t-1}
[dx,dv]
Wo
O = Wo·o_t
[B,n,H]